

KOCAELI UNIVERSITY
MECHATRONICS ENGINEERING



APPLIED MECHATRONICS ENGINEERING DESIGN
FINAL REPORT

PROJECT TITLE: Antenna Tracker System

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ABSTRACT

In this project, the focus is on the design and development of the Antenna Tracking System (ATS), which provides high-efficiency communication between the ground-aircraft or aircraft-satellite in unmanned aerial vehicles, enabling the alignment of directional antennas to the determined target. Since ATSs serve many special purposes such as satellite tracking, radar systems, rocket tracking, air traffic control, it is not possible to design and develop an ATS that can meet all demands. The project work was started by determining the needs of the Sub-Cloud Unmanned Aerial Vehicles (BIHA), designed within the Kocaeli University Autonomous Systems Technologies Team (KOUSTECH), to meet the communication needs of the projects for which the team provides solutions and, in the competitions, attended by the team. For this reason, the ATS designed and developed within the scope of this project was basically developed to meet the needs of BIHA. Developed because of needs, ATS is designed to track a BIHA passing at a height of at least 15m at a speed of 17m/s on a GPS basis, using telemetry data from BIHA and data generated on it. The high-altitude limitation depends entirely on the characteristics of the preferred antenna. ATS includes a dynamic model of the mechanism in the control loop to direct antennas with different weights and centers of gravity, and different antennas can be integrated into the system by simply changing some parameters. Incorrect coordinate data, which is one of the disadvantages of GPS-based tracking, is reduced by using IMU data and data from different sensors. Considering the possibility that the desired sensitivity and speed could not be achieved at close range despite all the efforts, an IP camera was placed on the ATS, making it open to system developments so that the BIHA can be observed continuously during the flight and image processing-based tracking can be performed in the future in close range tracking.

Keywords: Antenna Tracker, UAV, GPS, Pan-tilt mechanisms

ÖZ

Bu projede, insansız hava araçlarında, yer – hava aracı veya hava aracı – uydu arasında yüksek verimli haberleşmeyi sağlayan yönlü antenlerin belirlenen hedefe hizalanmasını sağlayan Anten Takip Sistemi (ATS)’in tasarımına ve geliştirilmesine odaklanılmıştır. ATS ler uydu takibi, radar sistemleri, roket takibi, hava trafiği kontrolü gibi birçok özel amaca hizmet ettiğinden, tüm taleplere cevap verebilen bir ATS tasarımı ve geliştirilmesi mümkün değildir. Proje çalışmasına Kocaeli Üniversitesi Otonom Sistemler Teknolojileri Takımı (KOUSTECH) bünyesinde tasarlanan Bulut altı İnsansız Hava Araçlarının (BİHA) ’ların, çözüm sunduğu projeler ve takımın katıldığı yarışmalardaki haberleşme ihtiyaçlarını karşılayacak şekilde ihtiyaçlar belirlenerek başlanılmıştır. Bu nedenle bu proje kapsamında tasarlanıp geliştirilen ATS temel olarak BİHA ihtiyaçlarını karşılamak için geliştirilmiştir. İhtiyaçlar sonucu geliştirilen ATS, BİHA dan gelen telemetri verileri ve kendi üzerinde üretilen verileri kullanarak en az 15m yükseklikten 17m/s hızla geçen bir BİHA yı GPS tabanlı takip edebilecek şekilde tasarlanmıştır. Yüksek irtifa sınırlaması tamamıyla tercih edilen antenin özelliklerine bağlıdır. ATS farklı ağırlıklardaki ve ağırlık merkezlerine sahip antenleri yönlendirebilmek için kontrol döngüsü içerisinde mekanizmanın dinamik modelini barındırmaktadır ve sadece bazı parametreler değiştirilerek farklı antenler sisteme entegre edilebilmektedir. GPS tabanlı takibin olumsuzluklarından olan hatalı koordinat verileri ise IMU verileri ve farklı sensörlerden gelen veriler kullanılarak hata oranı düşürülmüştür. Yapılan tüm çalışmalara rağmen yakın mesafede istenilen hassasiyetin ve hızın yakalanamaması ihtimali göz önünde bulundurularak ATS üzerine bir IP kamera yerleştirilip BİHA nın uçuş sırasında sürekli olarak gözlemlenebilmesi ve gelecekte yakın mesafe takipte görüntü işleme tabanlı takip gerçekleştirilebilecek şekilde sistem geliştirmelere açık hale getirilmiştir.

Anahtar Kelimeler: Anten Takip Sistemi, İHA, GPS, Pan-tilt mekanizması

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LIST OF ABBREVIATIONS

UAV: Unmanned Aerial Vehicle

IMU: Integrated Measurement Unit

GCS: Ground Control Station

ATS: Antenna Tracking System

GPS: Global Pointing System

MCU: Microcontroller Unit

CHAPTER 1

INTRODUCTION

With the development of sensor technologies, processing capabilities of microprocessors, battery efficiency, and electronic components, unmanned vehicles have found more applications in civil, commercial, and military fields. This study will focus on Unmanned Aerial Vehicles (UAV). UAVs can be controlled autonomously or manually, they have to provide data exchange with a station, this station is called Ground Control Station (GCS). All mission definitions and controls of unmanned vehicles are carried out through GCS [1]. With the development of mission capabilities, the bandwidth of the data produced in the vehicles and the data to be transferred from GCS have increased. Increasing data sizes require more optimized communication systems.

Two kinds of antennas with different characteristics are needed for communication, these antennas are called as directional and omni-directional. The main difference between these two antenna types is the radiation patterns. The radiation pattern in directional antennas has a certain angle while the radiation pattern covers all directions in omnidirectional antennas [2]. UAVs have omni-directional antennas on them because of they are constantly moving, energy consumption and platform weights are critical. On the other hand, directional antennas are used in GCSs cause of energy efficiency and weight are not critical [1]. For directional antennas to work, the target signal source must be kept within the beamwidth [2]. The tracking of the target signal source is performed by the antenna tracking system (Figure 1.1) by selecting the appropriate tracking method.



Figure 1.1: Digital Datalink Antenna Tracker [2]

1.2 Commercial Products

1.2.1 Octopus Tracking Portable Antenna (Figure 1.2)

By using GPS data from manned and unmanned aerial vehicles and GPS data of antenna tracking system, azimuth and elevation angles are produced and tracking is performed. With the ability to move between 360° on the azimuth axis and -13° - 96° on the elevation axis, it can exchange data at 12Mbps up to 115 km with 37 dBm gain. It has an automatic switching structure that enables the use of omni-directional antennas at close distances and directional antennas at long distances. The system communication speed can reach up to 12Mbps, the data going to GCS was transmitted using ethernet cables.



Figure 1.2: OCTAPUS Portable Antenna Tracker [3]

1.2.2 Digital Datalink Antenna Tracker (Figure 1.3)

It can move 360° on the azimuth axis and 90° on the elevation axis by using a GPS-based tracking method. The antenna weight it can carry can vary between 5-20kg. This change in weight also changes the maximum rotation speed of the antenna. It is resistant to bad weather conditions. It is the mobility that can be installed by one person in two minutes. It has the feature of self-calibrating [4].



Figure 1.3: Digital Datalink Antenna Tracker [4]

1.2.3 UKR Spec Systems (Figure 1.4)

It has high mobility on axes. Azimuth angle is 360° , tilt angle is between -40° - 75° . It has a built-in camera. It is resistant to high temperature differences. It has the possibility of working with a battery. There are two types, one is designed for longer range and the other is designed for easier transport [5].



Figure 1.4: UKR Spec Systems Antenna Tracker

Table 1.1: Comparison of commercial antennas

Product	Azimuth	Tilt	Method	Range (km)	Data Rates (Mbps)	Redundancy	Weight(kg)
OCTOPUS	360°	-13°-96°	GPS	115	12	Yes	10
DDL	360°	90°	GPS	100	25	No	5-20
UKR	360°	-40°-75°	GPS	150	10	Yes	31.4

1.3 STATE-OF-THE ART ON RELATED TECHNOLOGIES

Antenna tracking is carried out in two ways, electronic and mechanical tracking. In the electronic method, the antennas are placed in the form of a matrix (Figure 1.5) and the signal levels are measured and the communication is carried out by phase shifting the antenna with the highest signal level. Although this method is advantageous in terms of fast tracking, low maintenance and cost, low gain antennas cannot be used in this system.



Figure 1.5: Electronic method for tracking [6]

In the mechanical method, it is tried to keep the target signal source within the beam width by using the target antenna monopulse or the GPS positions of the aircraft where the target antenna is located, by moving the antenna in the designed degree of freedom. Although this method has problems such as speed limitations, being affected by ambient conditions, and the need for maintenance, it is one of the

advantages of the method that the antennas that can be used are redundant and that low-gain antennas can also be used [7].

Mechanical tracking method is divided into two parts as GPS-based tracking and monopulse tracking. In GPS-based tracking, the latitude, longitude, and height data of the Antenna Tracking System (ATS) and the latitude, longitude and height data of the UAV are used to determine the difference angles in the pan and tilt axes (Figure 1.6) with the help of mathematical functions.

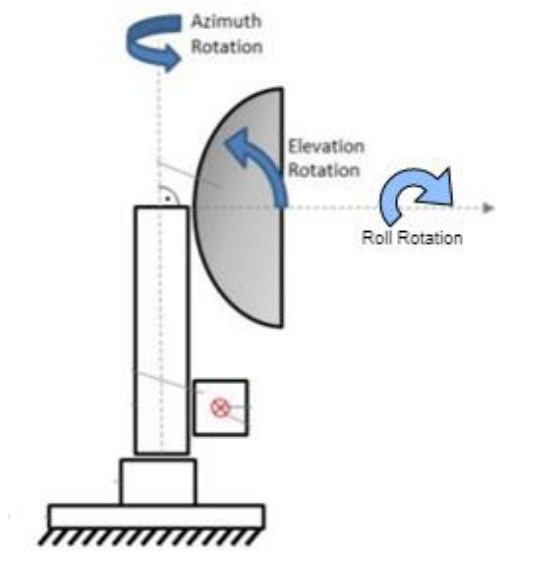


Figure 1.6: Axis on antenna [8]

The determined difference angles are processed in control algorithms and motion commands are sent to the motors [1]. In monopulse-based tracking, auxiliary antennas are used to determine the orientation in addition to the main antenna (Figure 1.7).

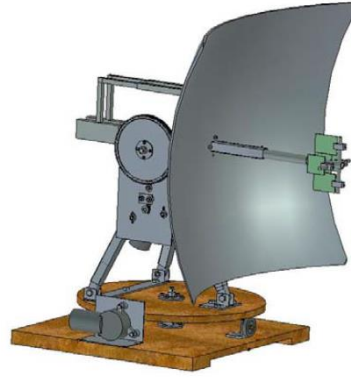


Figure 1.7: Application of monopulse tracking [9]

When the omni-directional antenna on the UAV aligns the tracking antenna exactly at 0° , equal signal quality is obtained on the auxiliary antennas, which indicates that the antennas are fully aligned. When any angular imbalance occurs in the pan and/or tilt axis, signal quality imbalance occurs in the auxiliary antennas. Signal quality is amplified with op-amp circuits and applied as analog input to MCUs, measured values from analog input are given as input to the control algorithm, and antenna tracking is performed by subtracting the necessary motor angles from the control algorithm (Figure 1.8) [9].

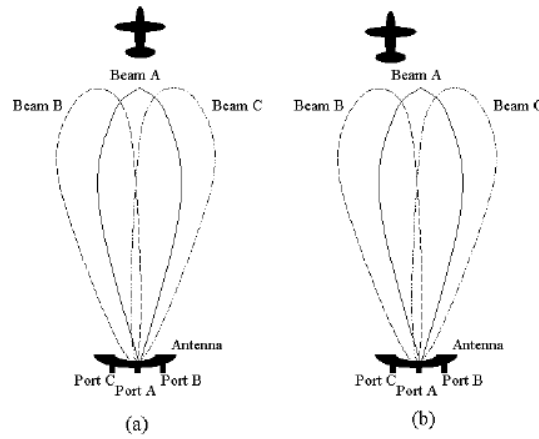


Figure 1.8: Monopulse Tracking Technique [9]

The mechanical capabilities of the Antenna Tracking System vary according to the characteristics of the system to be followed. It is necessary to provide orientation by processing the GPS data coming from the antenna and converting them to motor angles, systems with different degrees of freedom have been designed to achieve this. The most common system in the examined studies is the system with 2 degrees of freedom, known as the pan-tilt mechanism. Another system is the parallel

robot application known as Stewart Platform (Figure 1.9) with 6 degrees of freedom. Although the Stewart platform has a high load carrying capacity [10], it was kept in the background during the literature review due to its limited mobility in the pitch axis.

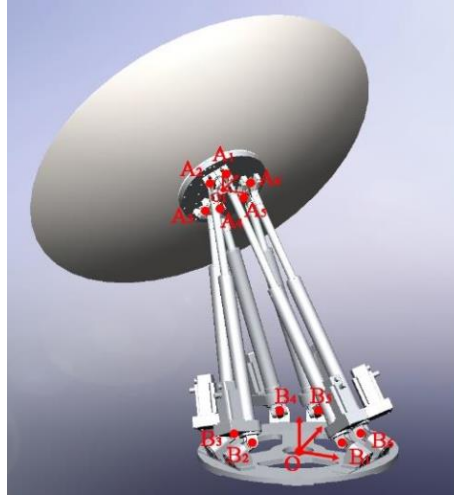


Figure 1.9: Stewart platform [9]

In the studies examined for the pan-tilt mechanism (Figure 1.10), gear mechanisms are used in all of them to transfer the torque required for antenna movement from the motors. Antenna and tilt mechanism are placed on a pedestal to provide 360° movement in the azimuth axis. Thus, an uninterrupted movement in the azimuth axis is ensured. Movements in the axes can be achieved by using different kinds of motors. These motors can be dc motor, brushless dc motor, brushless ac motor, stepper motor and dc servo motor [7].

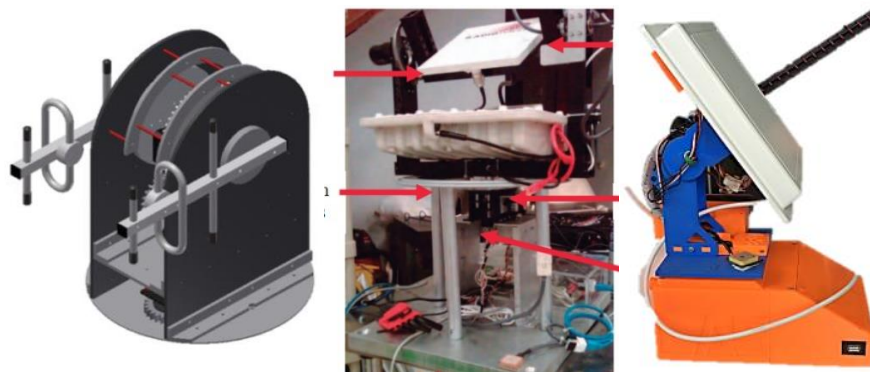


Figure 1.10: Pan-Tilt Mechanisms

a)Tilt axis mechanism [1]. b)Pan and Tilt axis mechanism [11]. c)Antenna Tracker Mechanism[1].

Due to their structure, mechanical tracking-based antenna tracking systems should follow $1/10$ of the beamwidth precision in order to achieve the most perfect data exchange. This requirement requires very precise movements at long distances, it requires more aggressive movements at short distances. Expecting fast and precise movement to be in different conditions in a system requires dynamically changing the control algorithm used. In order to realize this process, closed loop control is applied at the points where precise movement is required in some applications, while open loop control is used at the points where fast movement is required. In some studies, due to the high oscillations of traditional PI controllers at high speeds, system reliability has been increased by using LQG and H^∞ controllers in addition to traditional PI control [7].

CHAPTER 2

DEFINITION OF THE PROBLEM

Antenna tracking systems are frequently used in the communication of unmanned aerial vehicles with the GCS. Antenna tracking must be done precisely for continuous and large bandwidth data transfer. It is very important that the mechanism that will perform the tracking process can take the necessary positions according to the position of the unmanned aerial vehicle. In addition, the control algorithms to be applied and the activation of backup antennas when necessary will directly affect the stability of the system.

In this project, an antenna tracking system is designed for continuous communication of unmanned aerial vehicles. This antenna tracking system should be model-based and modular.

2.1 Project Requirements

- The system should have 360-degree tracking capability.
- The mechanism of system should have at least 2 degrees of freedom.
- The system should move to maintain a 15-degree beamwidth angle.
- The system should be weather resistant.
- It should be model based.
- It should have a modular structure.

2.2 Specifications for the Model

- Model is to be built on a 1:1 scale

2.3 Design Criteria

- Mechanism capabilities (%25)
- Sensor reading (%20)
- Motor driving (%20)
- Accuracy (%20)
- Tracking technique (%15)

CHAPTER 3

PROJECT PLANNING

In the organization of our project, the duties assigned to each team member were fulfilled. The distribution of tasks based on talent and interest has been carried out equally. Related sections and tables are presented.

3.1 ORGANIZATION OF THE PROJECT

Since this project was carried out by two people, task sharing was done in two areas to keep the efficiency at the highest level. Within the scope of the project, ÇAKMAK Mechanical will take part in the field of YÖNEY control.

3.2 EXPLANATION OF THE TASKS AND THEIR DEFINITIONS

3.1.1 Literature Survey (ALL TEAM)

Since the subject of the project offers specific solutions in the field of communication, it is necessary to determine the system architecture by making a comprehensive literature review, for this reason, approximately 1.5 months were allocated to the literature review at the first stage.

1.a Introduction (YÖNEY)

In this section, brief information is given about the problem itself, previous studies, and the subject to be focused on in the project.

1.b Commercial Products (ÇAKMAK)

Commercial antenna tracking systems have been researched and their operating parameters and properties are presented in tables.

1.c State of The Art on Related Technologies (ALL TEAM)

In this section, the historical development of antenna tracking systems and the solutions developed are investigated. The results found were compared.

3.1.2 Problem Definition (ALL TEAM)

The purpose of the tracking system was discussed and reported.

3.1.3 Project Planning (ALL TEAM)

The plan of the project is created in this section and the time table (Table 1) and Gantt charts (Table 2) are given below.

3.1.4 Conceptual Design (ALL TEAM)

4.a Project Requirements (ALL TEAM)

The needs determined by the source of the problem and the information in the literature search were also determined.

4.b Design Criteria (ALL TEAM)

The design criteria found in line with the project requirements were determined as a percentage in order of importance.

4.c Predesign (ALL TEAM)

Among the solutions encountered, the methods that are thought to solve the problem in the most efficient way have been carried out in a scalable way for analysis.

3.1.5 Calculations (ALL TEAM)

Preliminary design work sized considering payload and necessary calculations for motor selection, necessary calculations were made for the sensitivity and capture rate of the tracking system.

3.1.6 Part Selection and Ordering (ALL TEAM)

Necessary materials were determined by considering the project requirements and calculations. Cost and supply problems that may occur during the determination of these materials are also taken into account.

3.1.7 Detailed Design (ALL TEAM)

7.a Mechanical System (ÇAKMAK)

The mechanism selected in the preliminary design was designed in real dimensions in the SolidWorks environment and analyses were carried out on the ADAMS program.

7.b Control System (YÖNEY)

The SITL of the chosen tracking method was optimized by simulating the driving method of the selected motor, writing the code and performing the necessary estimation algorithms in the MATLAB environment. As a result of these studies, the outputs form the control system.

7.c Communication System (ALL TEAM)

According to the definition of the problem, a multidisciplinary study should be carried out in order to produce a solution. Some of this work required the system used for communication to work in an optimized way.

3.1.8 Manufacturing

8.a Production (ALL TEAM)

The production of the parts designed in the Solidworks environment was carried out in the industry.

8.b Assembly (ALL TEAM)

The assembly process of the produced parts was carried out.

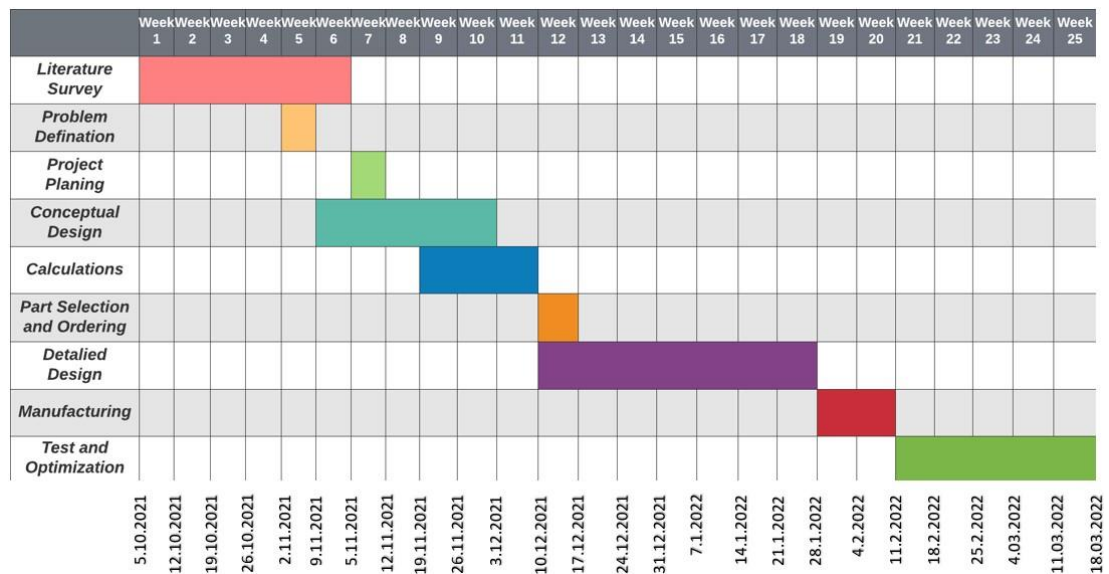
3.1.9 Testing and Optimization

The outputs of the produced system have been tested. These outputs were compared with the simulation results. The desired performance evaluation was made at this stage.

Table 3.2: Time Table

TASK ID	DURATION (#WEEKS)	TEAM MEMBERS
Literature Survey	6 weeks	ALL TEAM
Problem Defination	1 weeks	ALL TEAM
Project Planing	1 weeks	ALL TEAM
Predesign	5 weeks	ALL TEAM
Calculations	2 weeks	ALL TEAM
Part Selection and Ordering	1 weeks	ALL TEAM
Detailed Design	7 weeks	ALL TEAM
Manufacturing	2 weeks	ALL TEAM
Test and Optimization	5 weeks	ALL TEAM

Table 3.3: Gannt Chart



CHAPTER 4

CONCEPTUAL DESIGN

With the determination of the project requirements and the completion of the literature review, the necessary knowledge has been accumulated on how to meet each requirement. In line with the information obtained, design criteria such as the mechanical working principle of the ATS mechanism as a serial mechanism with 2 degrees of freedom as shown in figure 4.1 below, and the realization of GPS-based tracking were determined. The preliminary design and its criteria, which are formed because of the determined criteria, are detailed below.

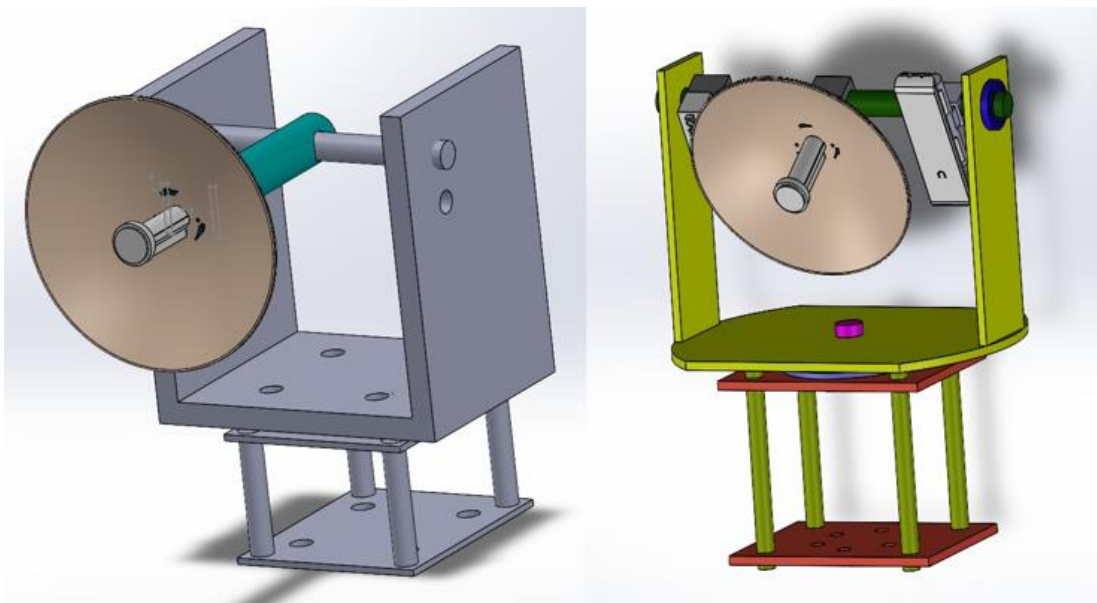


Figure 4.1: Evolution of our design

4.2 CONCEPT DEVELOPMENT AND PRESENTATION

We determined all the criteria we encountered during the design and listed the different results that could provide solutions to these criteria in the morphological chart given in Figure 4.2.

Part	Alternatives		
<u>Movement Mechanism</u>	 BELT AND PULLEY	 GEAR BOX	
<u>Electrical Source</u>	 BATTERY	 PLUG	 GENERATOR
<u>Drive System</u>	 SERVO	 STEP	 BLDC
<u>Controller Selection</u>	 ARDUINO	 PIC	 STM32
<u>GPS Selection</u>	 HERE-3	 HERE-2	 HERE-RTK
<u>Encoders</u>	 MECHANIC ENCODER	 OPTIC ENCODER	 MAGNETIC ENCODER
<u>Camera</u>	 IP CAMERA	 USB CAMERA	

Figure 4.2: Morphological chart of our design [12][13][14] ... [30]

4.3 EVALUATION OF CONCEPTS

This section sets the best options for systems. It was chosen according to important criteria such as ease of application.

- **Movement Mechanism: Gear Box**

Its efficiency is very high. It can be used even for low speeds. It can transmit high torque values. It is compact in construction [31].

- **Electrical Source: Battery and Plug**

Since the ATS will operate in different working environments, it is not always possible to connect to the electricity grid. Therefore, it is necessary to connect the battery to meet the planned operating time and to perform the charging process.

- **Drive System: BLDC**

Since BLDC motors have different driving methods according to the area of use, the system can be quickly adapted to different conditions by changing the code. The motor is planned to be driven with torque control.

- **Controller Selection: STM32**

STM's multi-processor models will be used for more effective operation of the created tracking and control algorithms.

- **GPS Selection: Here-3**

The Here-3 provides high resolution position reading through sensors such as IMU, barometer and navigation [32].

- **Encoders: Magnetic Encoder**

Since motion is detected by contact in mechanical encoders, the sensitivity of the encoders decreases during long operating times. Therefore, motion should be measured with non-contact methods for longer operating life. In this project, magnetic encoder was preferred due to its ease of use and high sensitivity [33].

- **Camera: IP Camera**

Since the communication within the system will be via Ethernet protocol, IP camera is preferred so that a converter is not used in image transfer.

4.4 BEST CONCEPT

The best concept was defined and objectively verified. Using the morphological chart, we identified sub-systems, their alternative components and application techniques. As a result, we tried to find the best solution for our original concept.

SUMMARY AND CONCLUSION

All the specified elements have been determined in the most appropriate way with the outputs obtained as a result of the literature review and market research. Once the design is completed in the SOLIDWORKS environment, it will be transferred to the ADAMS environment, and the control algorithm written in the MATLAB environment will be communicated with ADAMS, and dynamic simulation tests will be performed with the real flight data of the mechanism. During the simulations, link lengths and joint points will be continuously optimized, thus revealing the most efficient mechanism. With the optimization processes reaching a certain level, the selection of the BLDC motor will be completed. After the BLDC motor is selected and supplied, Hardware in the loop tests will be performed on the real mechanism of the embedded codes proceeding in parallel, and after it is seen that the system has reached.

REFERENCES

- [1] I. Changoluisa, J. Barzallo, J. Pantoja, S. Cayo, D. V. Navarro-Mendez, P. J. Cruz. "A Portable UAV Tracking System for Communications and Video Transmission.", *IEEE 4th Colombian Conference on Automatic Control*, 2019, pp. 1-2.
- [2] G. Nugroho, D. Dectaviansyah. "Design, manufacture and performance analysis of an automatic antenna tracker for an unmanned aerial vehicle (UAV).", *Journal of Mechatronics, Electrical Power, and Vehicular Technology*, vol. 9, pp. 32-40, 2018
- [3] OCTOPUS TRACKING PORTABLE ANTENNA,
<https://uavfactory.com/storage/ee040af5a9d88e954106c4f186a1c2b5235e5675.pdf>,
[Visiting Date: 19.10.2021]
- [4] DDL ANTENNA TRACKER, <https://shop.airbornedrones.co/products/antenna-tracker>,
[Visiting Date: 19.10.2021]
- [5] UKR SPEC. SYSTEMS, <https://ukrspecsystems.com/tracking-antenna-system-uav>,
[Visiting Date: 19.10.2021]
- [6] F. Tiezzi, J. Padilla, C. Vigano. "S-band Transmit/Receive Antenna with Electronically Switched Beams for Mobile Satellite Systems.", *6th Advanced Satellite Multimedia Systems Conference*, p. 64, 2012
- [7] A.A. Mulla, P.N. Vasambekar. "Overview on the development and applications of antenna control systems.", *Annual Reviews in Control*, vol. 41, pp. 47-57, Apr. 2016
- [8] Technische Universität München. "Calculation of Azimuth, Elevation and Polarization for non-horizontal aligned Antennas" Technical Document TD-1205-a, pp. 22, 11.02.2019
- [9] S. Jenvey, J. Gustafsson, F. Henriksson. "A PORTABLE MONOPULSE TRACKING ANTENNA FOR UAV COMMUNICATIONS.", *22nd International Unmanned Air Vehicle Systems Conference*, 2007, pp. 1-2.
- [10] Y. He, W. Li, Y. Wu, J. Wu, Z. Cheng. "A 6-DOF SHIP-BORNE ANTENNA PLATFORM WITH LARGE ORIENTATION WORKSPACE.", *39th International Conference on Ocean, Offshore and Arctic Engineering*, 2020
- [11] D. Díaz, M. Montilla, S. Suddarth. "Active tracking position antenna base a low-cost approximation with servo gimbals.", *IX Latin American Robotics Symposium and IEEE Colombian Conference on Automatic Control*, 2011
- [12] <https://shopdelta.eu/ip-camera-ds-2cd1023g0-i2-8mm-1080p-ikvision-l2-p13354.html>,
[Visiting date: 16.12.2021]

- [13] <https://www.morhipo.com/arc-arc-7200-1-3-mp-720p-mikrofonlu-usb-webc/33517832/detay>, [Visiting date: 16.12.2021]
- [14] <https://turkish.alibaba.com/product-detail/wide-review-3-7v-li-ion-18650-battery-2200mah-lion-battery-60680829180.html>, [Visiting date: 16.12.2021]
- [15] <https://www.elektrix.com/1x16-kaucuk-duz-fis>, [Visiting date: 16.12.2021]
- [16] <https://shop.aksa.com.tr/urun/aksa-aap-2200i-portatif-jenerator>,
[Visiting date: 16.12.2021]
- [17] <https://market.samm.com/arduino-uno-r3-orijinal>, [Visiting date: 16.12.2021]
- [18] <https://www.hepsiburada.com/pixhawk-here-3-can-gps-gnss-with-m8p-pm-HBC0000052PC3>, [Visiting date: 16.12.2021]
- [19] https://docs.px4.io/v1.9.0/en/gps_compass/rtk_gps_hex_here2.html,
[Visiting date: 16.12.2021]
- [20] <https://www.oyuncakhobi.com/here-rtk-gnss-gps-kit-m8p>, [Visiting date: 16.12.2021]
- [21] <https://www.robokolik.com/products/rotary-encoder-modulu>, [Visiting date: 16.12.2021]
- [22] <https://www.hamamatsu.com/us/en/product/optical-sensors/photo-ic/photo-ic-for-encoder/index.html>, [Visiting date: 16.12.2021]
- [23] <https://store.fut-electronics.com/products/magnetic-encoder-rotary-position-sensor-as5600>, [Visiting date: 16.12.2021]

- [24] <https://www.shreeharimanufacturer.com>, [Visiting date: 16.12.2021]
- [25] <https://www.turbosquid.com/3d-models/gear-mechanism-v-5-3d-model/1107093>,
[Visiting date: 16.12.2021]
- [26] <https://www.motorobit.com/urun/ds3225-25kg-su-gecirmez-dijital-servo-motor>,
[Visiting date: 16.12.2021]
- [27] <https://www.rhino3dprinter.com/urun/nema-17-step-motor-nema17>,
[Visiting date: 16.12.2021]
- [28] <https://turkish.alibaba.com/product-detail/light-weight-60kg-thrust-15470-30kw-bldc-motor-for-e-mega-drone-60766610527.html>, [Visiting date: 16.12.2021]
- [29] <https://www.microchip.com/en-us/product/PIC16F877A>, [Visiting date: 16.12.2021]
- [30] <https://gr.mouser.com/new/stmicroelectronics/stm-nucleo-development-boards/>,
[Visiting date: 16.12.2021]
- [31] <https://clubtechnical.com/gear>, [Visiting date: 16.12.2021]
- [32] <https://docs.cubepilot.org/user-guides/here-3/here-3-manual>, [Visiting date: 16.12.2021]
- [33] <https://www.akm.com/eu/en/products/rotation-angle-sensor/tutorial/type-mechanism-1/>
[Visiting date: 16.12.2021]